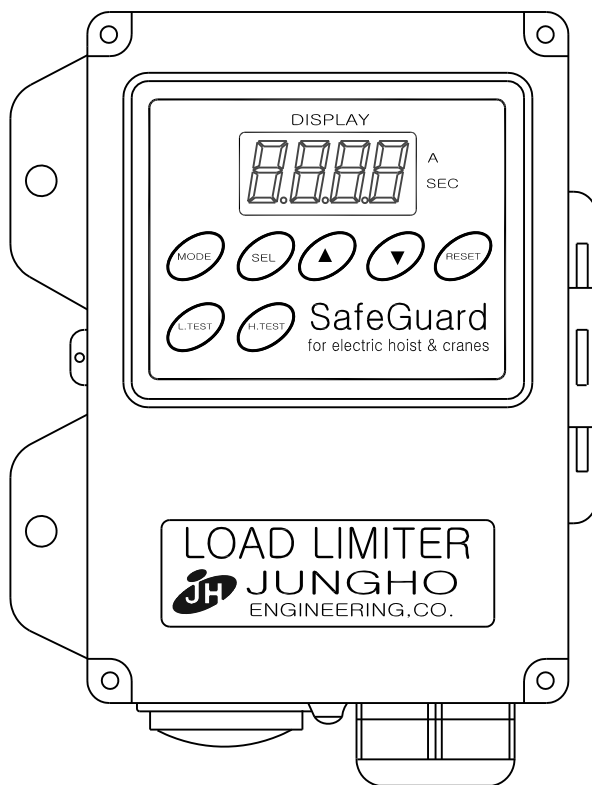


SPECIFICATION & INSTRUCTION

OVERLOAD LIMITER(PROTECTOR) for AC MOTOR

MODEL : JDL-100

JDLS-70



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WARNING

The authorized person or the person in charge of safety management only handle the load limiter .

This device is composed of the digital controlled circuits by using microcomputer & EEPROM. The microcomputer may have the possibility to be initialized by high level noise from the outside or by frequent switching "ON-OFF", and also may not be operated properly.

This matter will allow the motor to be overloaded and may cause the dangerous accidents relating to human life and injury. Therefore, the person in charge of safety management or the authorized workers at the circumstance exposed to frequent operation of power "ON-OFF" should regularly check these device. Especially, after power supplying "ON-OFF" to these devices, be sure to recheck the set values on display.

When initially installing the device, be sure to read carefully the instruction manual.

(When pushing the Up(▲) or Down(▼) keys one time during standby status, you can check the current set values in order.)



CAUTION

- 1) The inching operation exceeding 2 times per second causes error.
- 2) If the power source of load limiter is turned ON/OFF repeatedly, setting value for operation might be changed into the initialized status.
Be careful that this causes error or non operation.
Therefore, if the operating motor is turned ON/OFF frequently, the power source of load limiter should be wired to the previous circuit lines of power switch not to be affected by power switching ON/OFF.
(Refer to FIG1. the example of wiring diagram.)

1. INTRODUCTION & FEATURE

PATENT NO : 제0267456

PATENT NO : 제0338991

DESIGN PATENT NO : 제0240833

○INTRODUCTION

The Load Limiter is designed to fully protect AC motor against conditions which can cause damage. In addition to motor protection, the load limiter has feature that can protect overload mechanical equipment using AC motor such as HOIST, CRANE. If the motor of the associated mechanical equipment is overloaded, phase overcurrent is monitored through current transformer(CT) of the load limiter, the relay in the load limiter energize a contactor or breaker to cut-off power supply to motor under fault condition. The load limiter has a fine functions, high reliability, versatile application, each handling & operation.

○FEATURE

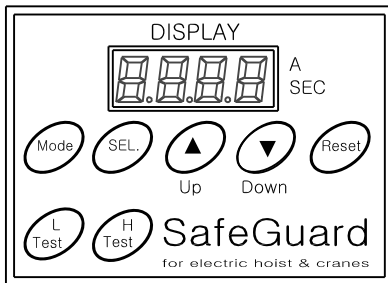
- 1) Since the load limiter having microcomputer built-in has the functions of delaying reset time & memorizing ascending time, it is automatically reset, if hoist or crane is descended during the reset time.
- 2) Fine setting time & current is available since the device is digital type.
- 3) In case of being used for creep hoist having creep speed & high speed, the device displays each trip signal & each current setting.
- 5) Running current of motor is displayed itself. Thus, setting current is easily done without extra current meter.
- 6) When overload, LED flashing & alarm sound as trip indication provide a clear distinction.
- 7) Wiring & installing are very simple by use of external current transformer.
- 8) Main control PCB & power PCB are easily separated for maintenance by use of each connector.
- 9) For easy maintenance, connection cable is easily separated from main body.
- 10) Check the inching number & total running time is available.
The device must be operated by authorized person only.

- 11) The value set is easily checked by pressing "Up" (▲) or "Down" (▼) Keys without removing the seal or opening its cover.
- 12) The device can be used as the over current relay for protecting AC motor.

2. SPECIFICATION

Model No.		JDLS-70	JDL-100	
APPLIED CT		"C" type	Add the subsidiary CT (ratio=250:5) whit "B" Type (in case of using the ratio of 500:5 wind two turns)	
APPLIED MOTOR	capacity	less than 99.9A	less than 99.9A	less than 250/500A
	voltage	AC 200V ~ 480V		
Setting range of load current		0.5 ~ 99.9A	0.5 ~ 99.9A	5.0 ~ 999A
Scale of current setting		0.1A (100mA)	0.1A (100mA)	1.0A
Adjusting range of start time		0.1 ~ 10 SEC		
Adjusting range of overload time		0.1 ~ 10 SEC		
Adjusting range of reset time		0.1 ~ 10 SEC		
Power source		AC24/48.48/110.110/220V $\pm 15\%$ 50/60HZ (by order)		
Capacity of relay contact		1a, 1b (Contacts), AC250V 5A (resistive load)		
Ambient temperature & humidity		-10~50°C, less than 80%		
Surge withstand voltage		AC1500V one minute duration		
Poewr consumption(MAX)		1.0VA (1.5VA)		

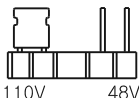
● DETAILS of DISPLAY WINDOW



LOAD LIMITER WINDOW

- (▲) Up (▼) Down keys :
Up "(▲)" or Down"(▼)" Keys are used for changing or checking the setting value.
- Reset Key :
This key is used for resetting relay after a trip when pressing the test keys and when the motor is overloaded.
- "L" "H" mode test :
Thes keys are used for testing the operation of high or low speed.

Input voltage selecting tap.



By moving tap,you can select input voltage.

※ The left Fig is for 110V.

3. Caution for Installation

3.1 Install Suggestion.

- 1) Please, mount the units horizontally or vertically.
- 2) Do supply the rated voltage. Ensure that the rated voltages 48v/110v or 110v/220v is supplied to the device, and that the tap for voltage selection is set as desired.
If not, the device may be burned out.
- 3) When the controller for AC motor is required for testing the surge voltage withstand capability on exceeding AC 1,500V, after removing the load limiter from it, the test should be carried out
- 4) Connecting wires of CT of the load limiter should be twisted and its length should not be exceed 1meter.

3.2 Checking & Maintenance

- 1) Check the setting value of the device before working everyday and also check that it is operated normally by pushing the test key.
 - 2) Ensure that the operating status of the device under being overload should be checked & tested.
- For periodic checkup, you can check frequently, the current setting value by pushing once the Up (▲) or the Down (▼) keys, not using the setting keys, under being in standby status and being sealed.

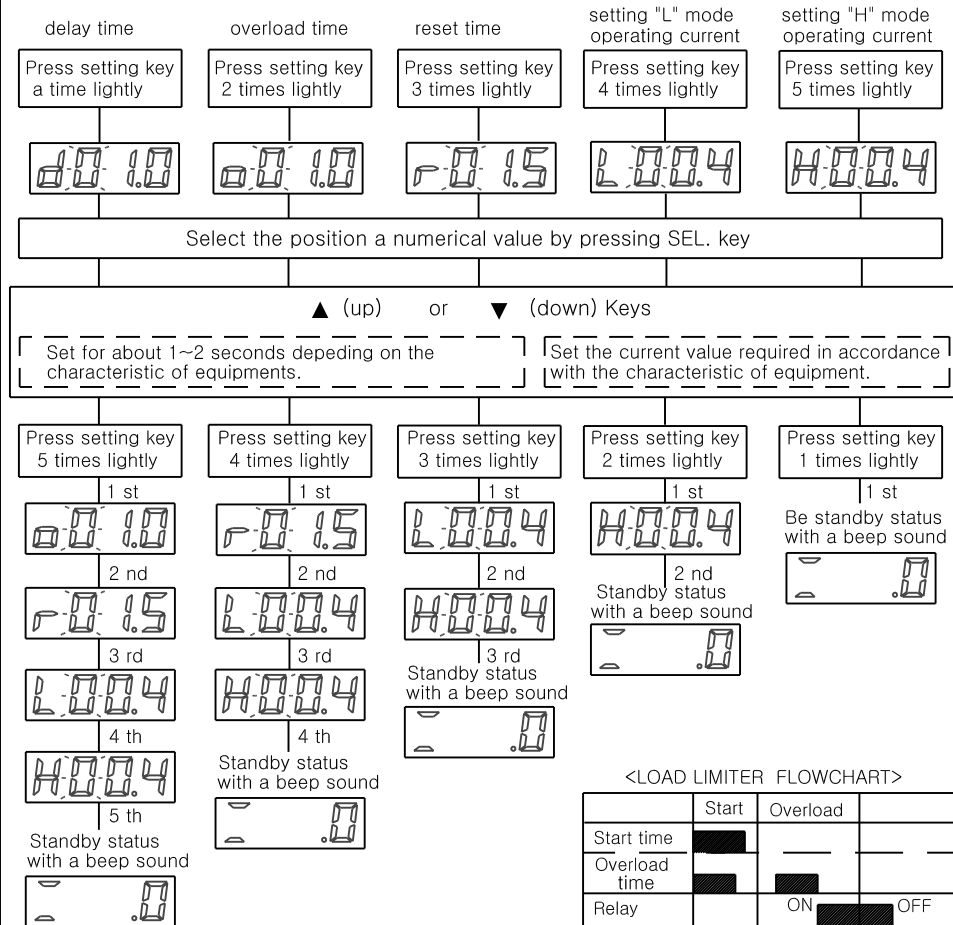
4. Functional setting range and unit

DISPLAY	Functions	Initial setting	Setting range	Setting unit
	"L" "H" start delay time	1.5 (Sec)	0.1~10 (Sec)	0.1(Sec)
	"L" "H" overload time	1.5 (Sec)	0.1~10 (Sec)	0.1(Sec)
	"L" "H" reset time	1.5 (Sec)	0.1~10 (Sec)	0.1(Sec)
	Setting current under low speed	Rated current value of motor	Depending on hoist model rating.	
	Setting current under high speed			

5. Display explanation

Display	Condition		Condition
	Poewr on		Overload status on operating
	Operating current of "L" mode		Overload operating of "L" mode
	Operating current of "H" mode		Overload operating of "H" mode
	"L" "H" starting delay times		Overload operating of "L" and "H" mode

6. Functional setting method



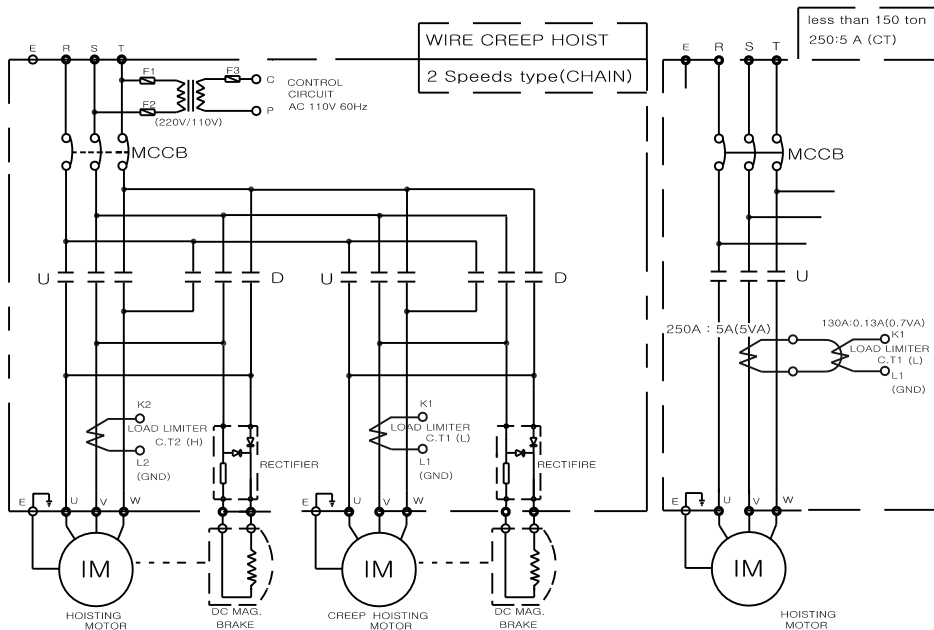
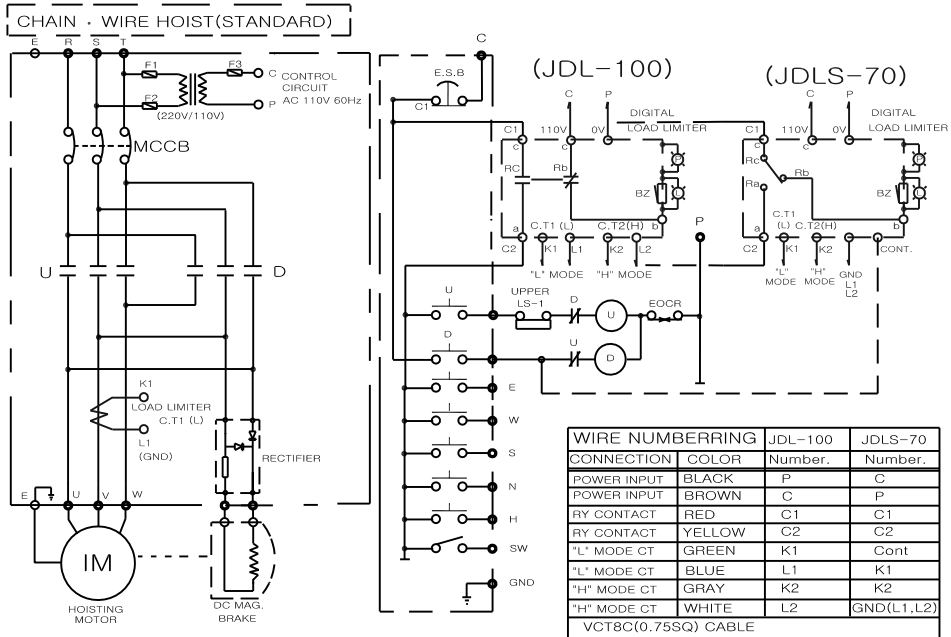
<REMARK>

- 1.Start delay time is initially set 1.5 second. Therefore you need to reset it again except in case of long start time.
- 2.It is safe that setting reset time is longer exceeding 0.2 second than start delay time.
- 3.In case of applying the single speed hoist, the current of "L" and "H" modes should be set indentially each other by sensing current from 2 lines.
At this time,sensing current is obtained from 2 phase power lines
- 4.In case of applying the two speed hoist (creep type, pole change type), "L" mode should be set from the current of low speed and "H" mode should be done from the current of high speed.
- 5.Overload time should be set shorter than start delay time to protect motor against overcurrent.
Setting within 1.5 second is normally recommended.

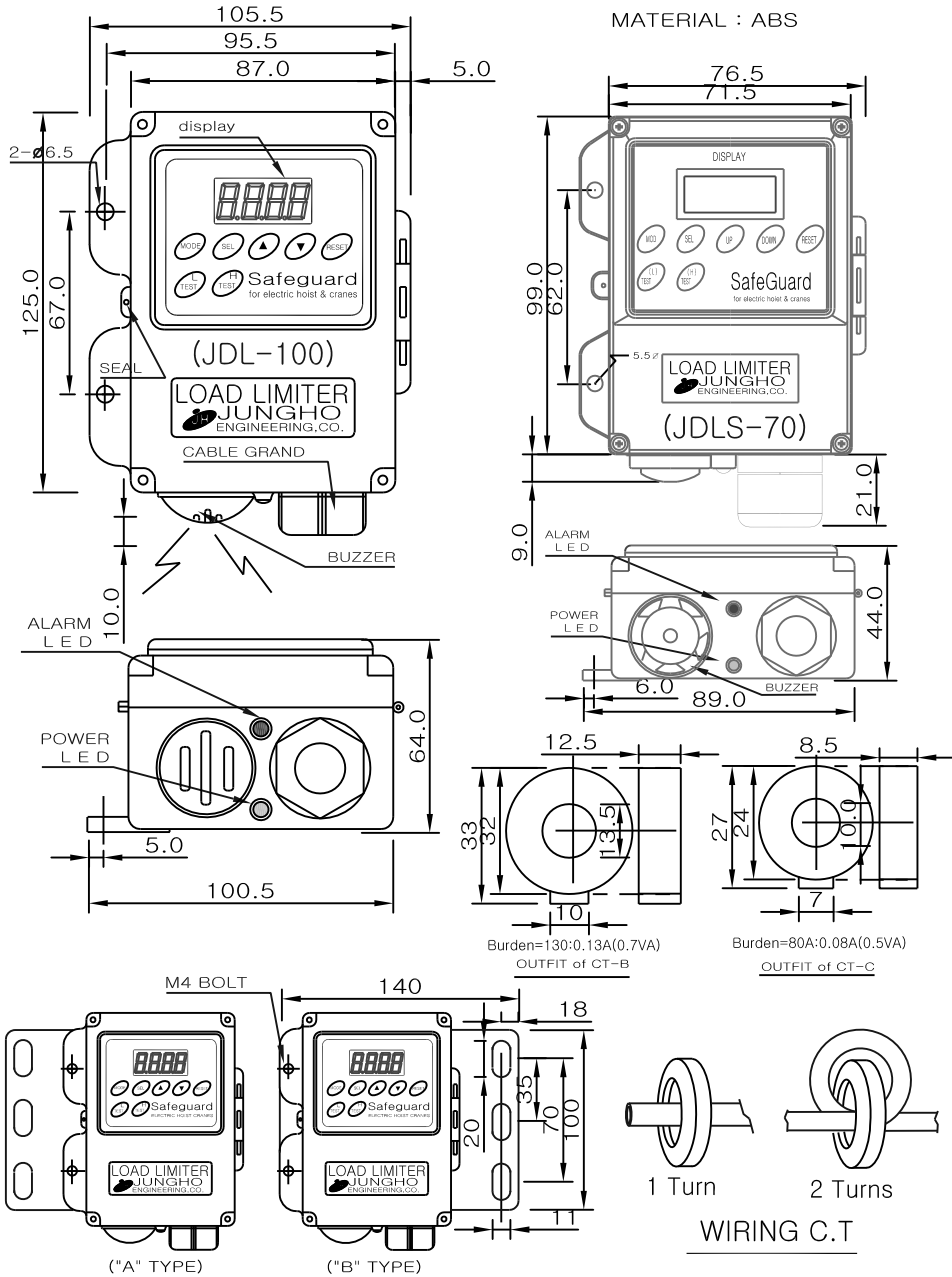
<LOAD LIMITER FLOWCHART>

	Start	Overload	
Start time			
Overload time			
Relay		ON	OFF
Buzzer/LED			
Up		Stop	
Down			
Reset time			

7.FIG1.EXAMPLE OF WIRE DIAGRAM(JDL-100,JDLS-70)



8.FIG2. MOUNTING OUTPUT(JDL-100,JDLS-70)



Load Limiter

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JUNGHO ENGINEERING, CO.

POSTAL CODE : 14335

#10-7, Abangri-3gill, Gwangmyeong-City,
GYEONGGI-DO, KOREA

TEL : +82-2-2681-0553

FAX : +82-2-2612-3295

E-MAIL : junghoeng@hanmail.net

www.junghoeng.com